

# Technical Approach

Intelligent Land Record Digitization & Validation System (SIH26018): technologies, where each is used, and the software architecture

## Approach in seven steps

|   |                                                                                                                                                                                                                                                                                                                   |
|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <b>Capture</b><br>Officer uploads a scanned land record (PDF, JPG or PNG, up to 20 MB). The file is stored with a SHA-256 fingerprint, so the same document uploaded twice is detected.                                                                                                                           |
| 2 | <b>Clean the image</b><br>Pages are straightened (skew up to $\pm 10^\circ$ ), yellowed paper, stains and shadows are removed, table grid lines are erased and faded ink is recovered (OpenCV). Clean pages skip this step.                                                                                       |
| 3 | <b>Read the text (OCR)</b><br>Tesseract 5 reads 12 languages (English, Hindi, Marathi, Punjabi, Bengali, Gujarati, Odia, Tamil, Telugu, Kannada, Malayalam, Urdu). Hindi uses our model fine-tuned on land-record text. PDFs with a text layer are read directly. OCR runs as background jobs; pages in parallel. |
| 4 | <b>Extract the details</b><br>13 fields (owner, father's name, khasra / survey no., khata no., village, tehsil, district, area, date, ...) are found by label matching in Hindi and English; optional AI checks the values against the page image.                                                                |
| 5 | <b>Score confidence</b><br>Every value gets a 0-100 score with reasons (how it was found, OCR word confidence, format checks). Each page reports the share of words read clearly, e.g. "5% read clearly" on a handwritten page, so officers know what to trust.                                                   |
| 6 | <b>Validate</b><br>Rule checks (required fields, formats, area range), cross-record conflicts (same parcel with different owners, duplicates, area and khata mismatches), Hindi-English name matching and cadastral map (GIS) checks: parcel on map, area and boundary vs map, overlaps.                          |
| 7 | <b>Review &amp; learn</b><br>Officers verify or reject each record. Their corrections are recorded to measure accuracy and correct repeated errors automatically, and typed-in text lines become training data for fine-tuning the OCR. Each parcel gets an ownership timeline (digital twin).                    |

## Design principles

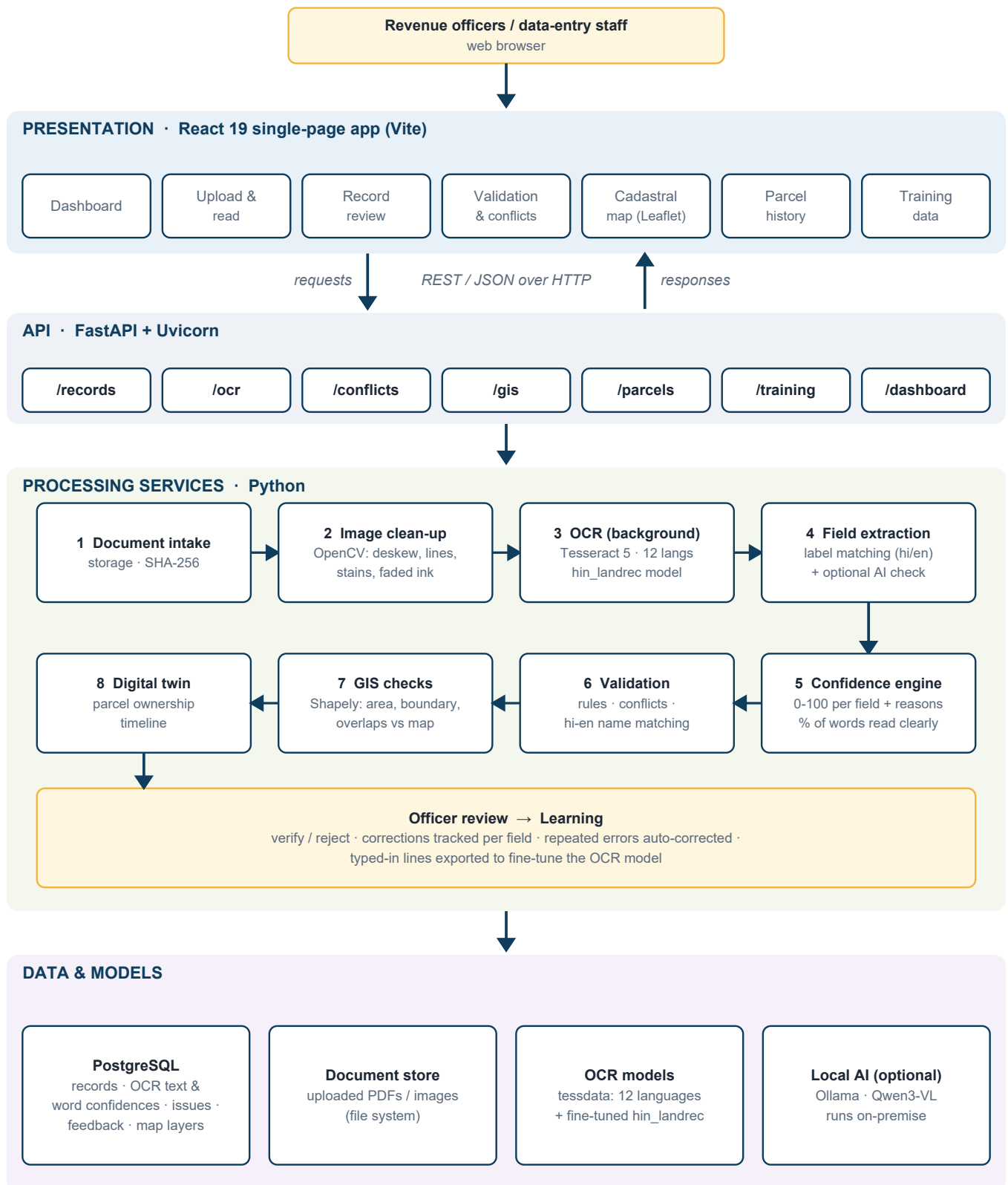
|                            |                                                                                            |
|----------------------------|--------------------------------------------------------------------------------------------|
| <b>Human in the loop</b>   | The system suggests; an officer confirms. Nothing is saved without review.                 |
| <b>Explainable</b>         | Every score and warning says why. Matching and validation are rule-based, not a black box. |
| <b>Honest about limits</b> | The share of text read clearly is always shown, including for handwritten pages.           |
| <b>Runs on-premise</b>     | OCR, AI checking (local model) and the database can all run inside the office network.     |
| <b>Improves with use</b>   | Officer corrections and transcriptions become data to fine-tune the OCR model.             |

## Technologies used and where they are used

| Layer                   | Technology                                                                        | Where it is used                                                                                                                            |
|-------------------------|-----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frontend</b>         | React 19, React Router 7                                                          | Officer web app: dashboard, upload & read, record review, validation queue, conflicts, map, parcel history, training pages                  |
|                         | Vite 7                                                                            | Development server (with API proxy) and production build of the web app                                                                     |
| <b>Maps</b>             | Leaflet 1.9 + OpenStreetMap                                                       | Cadastral map page and the parcel map on each record                                                                                        |
|                         | Leaflet-Geoman                                                                    | Drawing and editing a parcel boundary on the map                                                                                            |
| <b>Backend API</b>      | Python, FastAPI, Uvicorn                                                          | REST API: records, OCR, conflicts, GIS, parcels, training, dashboard                                                                        |
|                         | Background job queue (Python thread pool)                                         | OCR runs after upload without blocking; the page updates itself when reading finishes                                                       |
| <b>Database</b>         | PostgreSQL, SQLAlchemy 2, psycpg 3                                                | Records, OCR text and per-word confidences, validation issues, officer feedback, conflict reviews, map layers (GeoJSON)                     |
|                         | File storage + SHA-256                                                            | Uploaded documents on disk; fingerprint detects duplicate files                                                                             |
| <b>OCR</b>              | Tesseract 5 (LSTM) via pytesseract                                                | Reading scanned pages in 12 languages                                                                                                       |
|                         | Fine-tuned model hin_landrec (tesstrain)                                          | Hindi land-record text; character error rate 11.2% to 1.0% on held-out synthetic lines in unseen fonts                                      |
|                         | PyMuPDF                                                                           | Reading PDF text layers directly; rendering scanned PDF pages at 300 dpi                                                                    |
| <b>Image processing</b> | OpenCV, NumPy, Pillow                                                             | Deskew, background and stain removal, table-line removal, faded-ink thresholding, line cutting for training                                 |
| <b>Extraction</b>       | Label matching (Python regular expressions)                                       | 13 record fields from Hindi and English labels, Devanagari digit conversion                                                                 |
|                         | Optional AI: vision-language model (local Qwen3-VL via Ollama, or a cloud AI API) | Checks extracted values against the page image; the local option keeps data on-premise                                                      |
| <b>Validation</b>       | Rule engine (Python)                                                              | Required fields, formats, area range, numbers entered vs document text                                                                      |
|                         | difflib + phonetic key                                                            | Matching names written in Hindi and English (the same name in Devanagari and in English letters) for owner-conflict and duplicate detection |
| <b>GIS</b>              | Shapely, GeoJSON                                                                  | Parcel area, boundary vs cadastral map, overlapping boundaries                                                                              |
| <b>Learning</b>         | Feedback store + tesstrain export                                                 | Accuracy per field, automatic correction of repeated errors, training data for OCR fine-tuning                                              |
| <b>Configuration</b>    | python-dotenv (.env)                                                              | Database, OCR languages, AI on/off and provider, worker count                                                                               |

## Software architecture

A layered web application: a React front end talks to a FastAPI back end over REST. Documents flow through the processing services (numbered in order); results are stored in PostgreSQL and every value is confirmed by an officer.



## How a document flows through the system

|                            |                                                                                                                                                                                                       |
|----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Upload</b>              | React upload page → <b>POST /api/records/upload</b> ; file saved, record created, OCR job queued. The response comes back at once.                                                                    |
| <b>Read</b>                | Worker thread: PyMuPDF renders pages → OpenCV clean-up (only if needed) → Tesseract OCR, pages in parallel → text and per-word confidences saved.                                                     |
| <b>Extract &amp; score</b> | Label matching finds the fields; optional AI checks them against the image; the confidence engine scores each value and counts the words read clearly.                                                |
| <b>Validate</b>            | Rules, cross-record conflicts and GIS checks run; issues are stored with the record.                                                                                                                  |
| <b>Review</b>              | The record page polls until reading finishes, then shows the verdict, the details found (with "Looks right / Please check / Unsure"), the text coloured by confidence, and the document side by side. |
| <b>Decide &amp; learn</b>  | Officer copies values into the record, verifies or rejects; the feedback is stored for accuracy tracking and OCR fine-tuning.                                                                         |

## Measured results (for the slides)

|                             |                                                                                                                                                                            |
|-----------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Fine-tuned Hindi OCR</b> | Character error rate 11.2% → 1.0% on held-out synthetic land-record lines in fonts not used for training (stock vs fine-tuned Tesseract).                                  |
| <b>Image clean-up</b>       | Fields read correctly on degraded copies of a sample Khatauni: 10 of 54 → 43 of 54 (tilted, stained, faded and low-resolution scans).                                      |
| <b>Speed</b>                | Background jobs: upload returns in under 1 s. Parallel page OCR: 4-page scanned PDF 34.8 s → 18.2 s on a 12-core PC.                                                       |
| <b>Clean printed page</b>   | All 9 fields of the sample Khatauni found, about 92% OCR confidence, a few seconds.                                                                                        |
| <b>Handwritten pages</b>    | Printed-text OCR reads only about 5% of the words clearly; the system shows this figure and asks the officer to enter the details. Handwriting recognition is future work. |

Test figures come from the project's own measurements on synthetic and sample documents; results on real archive scans will differ.